

Operation manual

Cooling water circulator

A-OK600-1



- Please read this manual before your operation.
- Please ground the earth cable for prevention of electric shock
- Please store this operation manual



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Thank you for your purchase this unit!

Please read this operation manual for more safety and smooth operation and ensure your proper operation.

Mechanisms and specifications may be subject to change for improvement without notice.

Thank you for your understanding there might be any difference with this manual then.

Attention for safety

Following attentions are to use the product in safety and to obviate any damage.

■ Warning and notice



warning ▪ ▪ If you do wrong operation in defiance of this icon, it may be to deliver significant damage to human body.



notice ▪ ▪ If you do wrong operation in defiance of this icon, it may be to deliver any damage to human or mechanical body.

■ Meaning of icons



▪ ▪ This icon means to inform attention action.
The detail action is written near this icon.



▪ ▪ This icon means to inform prohibited action.
The detail action is written near this icon.



▪ ▪ This icon means to inform necessary action.
The detail action is written near this icon.



Must conform for safety



Warning

If you do wrong operation in defiance of this icon, it may be to deliver significant damage to human body.



Electrical power must keep over 15A at single phase 200VAC (Don't put many plugs in one outlet)

If the short-circuit protector work, please let PASCAL Co., Ltd. or seller know that.

If you try the forced return of the power supply, it may cause an electric shock, a fire or other troubles.

Don't install this circulator at the place where any flammable gas may leak

It may cause a fire or other troubles.

Please install this circulator at the level stable place.

It may cause a water leak, an electric shock, a fire, a breaking, an injury or other troubles by the rollover or the fall by the bad installation.



Grounding the earth cable

Please ground the earth cable completely. If it's not complete, it may cause an electric shock. There is the connection terminal for grounding at electrical power connection area

Don't use any flammable water

The liquid which you can use for the circulation is distilled water or clean water (not pure water or deionized water)

(If you want to use at under 5 degrees C, please use the ethylene glycol)

Don't use any flammable water. It may cause a fire.



Inhibition of shutoff pumping

If the circulator pump runs at shutoff condition, it may cause a pump failure or any unexpected trouble by break of hose or piping.



Notice

If you do wrong operation in defiance of this icon, it may be to deliver any damage to human or mechanical body.



Don't disassemble and modify

Don't disassemble and modify. It may cause an injury, a fire, an electric shock or any unexpected trouble by abnormal working.

Don't move this circulator in a state with water

It may cause a water leak, an electric shock or other trouble.

Don't get any impact

It may cause any unexpected break.

Don't dump water on this circulator

It may cause an electric shock, a fire or other trouble.



Attention to burn

The inside of this circulator may be high temperature. Don't touch the area except operation panel and water intake. If you touch the area, please glove to protect against the burn.



Attention to rolling in pump

Please be careful that any foreign substance is not in the pump inlet
It may cause a pump break.



Periodic confirmation to work short-circuit protector

It may cause an electric shock if you continue using it with break down and it doesn't work at a critical time

Confirmation of water level

Please confirm the water level to start running and please refill water if water level is under the limit.

It may cause a short lifetime if you continue running this circulator with low water level for a long time.

Summary of this circulator

This circulator can cool a heating point to circulate cooling water and keep the heating point at stable constant temperature.

Then this circulator can be used for constant temperature water tank which needs optional shelf plate.

Several parts name

Please refer outline drawing

Preparation for setup

- This circulator uses electronic components so please install it at clean place. Then don't use it at high-temperature and humidity, in dusty place or near caustic substance.

Connection of power cable

- Please ground the earth cable and don't put many plugs in one outlet.
- Please confirm that the primary power voltage is AC 200V +/-10% at 50/60Hz by a multimeter.

Hose connection and water preparation

- Please connect hoses between the heating point and this circulator. Please connect drain hose. Please be careful not to leak water.
- Please open the tank lid above this circulator and put water in tank up to 30mm from top. *(The water level will be low after this circulator runs because water move pump, hoses and heating unit. Then please put water in tank up to 30mm from top again after this circulator runs for several seconds.)*

Preparation to start running

- Please turn on the main power supply which is located at left side of this circulator and confirm that the temperature adjuster of the operation panel is turned on the power.
- Please confirm that the alarm lamp is turned off the light after power-on.
- Please confirm that temperature adjuster accurately displays PV which is current temperature.

Running operation



The current water temperature is displayed on PV/SV display panel when the power is supplied to this circulator and there are no error after warming-up time.

How to change the setting temperature

1. When you push [SP↑] button, the setting temperature (SP) is displayed on PV/SV display panel and the leftmost digit becomes a blinking state.
2. When you set the setting temperature, please change the number to push [SP↑] [MV↓] [RUN←] [M/En→] buttons. The blinking is moved to the right digit when you push [M/En→] button.
3. When the blinking is moved to the rightmost digit and you push [M/En→] button, the setting temperature is fixed and the display returns to the current water temperature.

How to run

This circulator starts the running when you push [RUN←] button over 2 seconds and release the button after the lamp above [RUN←] button is turned on.

This circulator stops the running when you push [RUN←] button and release the button after the lamp above [RUN←] button is turned off.

* The running stop is 2 seconds later after to push [RUN←] button.

When the normal running starts, the lamp above [RUN←] button is turned on.

After the running starts, the pump start working and then the refrigerator starts working.

* It takes several seconds that the refrigerator starts working but it's not failure. The refrigerator may not work immediately depend on the setting temperature. When all normal running starts, water temperature in tank is kept at a constant temperature by PID control.

This circulator has the terminal block for the remote running and starts the running when two RMT lines on the terminal block is shorted. In case of remote stopping, this circulator stops the running when this short on the terminal block is opened.

Alarm display

In the alarm status, PV/SV display shows the number from 0 to 13.

Safety circuit

This circulator has water level alarm, water level failure alarm, temperature low limit alarm, temperature upper limit alarm, refrigerator high pressure failure alarm, pump motor overcurrent alarm, refrigerator motor overcurrent alarm, flow late failure alarm for safety circuit

Alarm type	Status	Recovery action
Pump motor overcurrent	Stop running Error no. 1	Please keep stopping running for a while. Then please start running again after power turning off once or error reset.
Refrigerator motor overcurrent	Stop running Error no. 2	Please keep stopping running for a while. Then please start running again after power turning off once or error reset.
Refrigerator high pressure failure	Stop running Error no. 4	Please fix refrigerator high pressure failure (keep stopping running for a while) Then please start running again after power turning off once or error reset.
Water level	Keep running Error no. 6	The lamp will turn off automatically after water rising so please fill water until the lamp turns off.
Water level failure	Stop running Error no. 7	Please fill water until adequate water level. Then please start running again after power turning off once or error reset.
Flow late failure	Stop running Error no. 8	When circulating water can't flow normally, the circulator stops several seconds later after starting running. Please confirm clogged filter or shutoff pumping then fix the failure. And then please start running again after power turning off once or error reset.
Temperature upper limit	Stop running Error no. 11	Please cool temperature under alarm setting temperature. Then please start running again after power turning off once or error reset.
Temperature low limit alarm	Stop running Error no. 13	Please heat water temperature over alarm setting temperature Then please start running again after power turning off once or error reset.

* Please let PASCAL Co., Ltd. when the alarm lamp turns on. It may cause any unexpected trouble by repeating alarm and clear.

The way of the alarm reset is the power turning off so please turn off the power once after you fix the alarm cause.

Maintenance and check

Please keep water in tank clean

If you use water which is impure or bad water quality, the water may dirty the pump and inside of refrigerator. Then pumping ability and cooling power may fall down. And then this circulator may not have the specified performance.

Please exchange water with distillated water or clean water (not pure water or deionized water) routinely.

Please clean up the dirt on the surface with neutral detergent.

It might be damaged when I use surface hard cloth or scrubbing brush

Don't clean with cleaners such as an organic solvent or the acid

Long rest

If this circulator doesn't do running for a long time, please turn off the power, drain water, cover it to protect dirt and foreign substance, and set under the dry air.

The case of repair and question

Don't repair this circulator by yourselves.

When you require repair and question, please inform us following checks.

1. Does the power cable connect to the appropriate power line?
2. Does the short-circuit protector turn on?
3. Does [RUN←] button turn off after alarm reset?
4. Is there any shutoff pumping point in the water way?